Milestone 2 (Team 8)

Requirements

Project Description:

Cornell needs a single, reliable way to see who has access to which data in the Central Data Warehouse & Data Lakehouse and how that access is being used. Right now, people pull data from different systems and share spreadsheets a few times a year. Our team will build a proof-of-concept BI system (data model + ETL + dashboards) that centralizes security mappings and audit logs (Finance, Contributor Relations, Student) and turns them into dashboards for IT Security, University Audit, and Data Stewards.

Functional Requirements:

- The dashboard(s) for the project sponsors to access and use that display account data access and usage
- See access and usage across the warehouse/lakehouse for designated staff
    - Easily identify if an account owner changes their employment status, unit, or department, and when the data a change is made

Filter and group by:

- account (netID or application account)
- department and unit/college (account → department → unit)
- data mart, role, and table,
- account type (user vs application)
- employment status (Active / Inactive / Retired)
- time window (last 3 months by default)

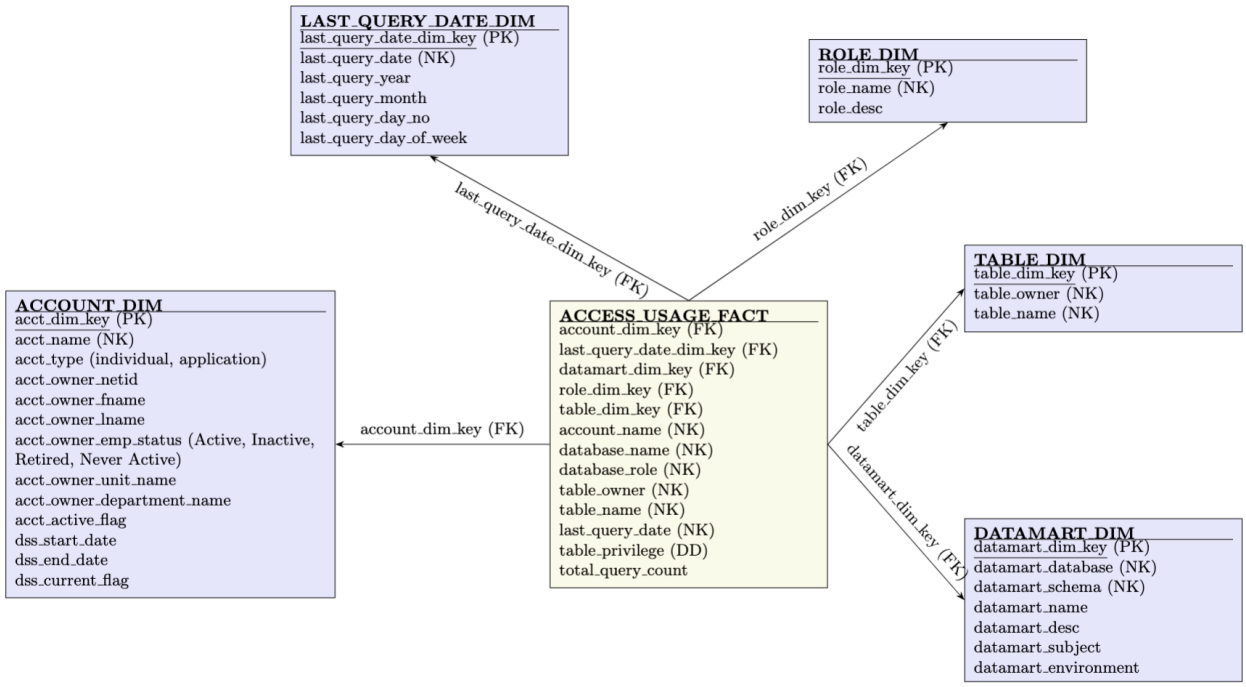
Show, for each account:

- What they can access (roles, tables).
- Type/level of access (SELECT, INSERT, DELETE).
- How much they used it (query counts in the last 3 months).
- When they last used it (by role & by table).

Dashboards:

- Which accounts shouldn't have access (retired/inactive)?
- Which accounts aren't using their assigned roles (no activity in 90 days)?
- User vs application account access and usage.
- Usage by role and last access by role.

Dimensional Data Model



table_privilege is a degenerate dimension (DD) in the fact table for simplicity.

Data Dictionary

Table Name	Column Name	Data Type	Business Description/ Definition
account_dim	account_dim_key (PK)	INTEGER	Unique surrogate key for the account in the dimension
account_dim	acct_name (NK)	STRING	Account name that can be either netid or application username?
account_dim	acct_type	STRING	Type of account (e.g. individual, application)
account_dim	acct_owner_fname	STRING	First name of the account holder
account_dim	acc_owner_lname	STRING	Last name of the account holder
account_dim	acct_owner_emp_status	STRING	Status of account (e.g. active, inactive, retired, never active)

account_dim	acct_owner_unit	STRING	The college/unit that the account belongs to
account_dim	acct_owner_department	STRING	The department that the account belongs to
account_dim	dss_start_date	DATE	Date when a particular version of the account record becomes effective or valid
account_dim	dss_end_date	DATE	Last date the version is valid or set to expire
account_dim	dss_current_flag	BOOLEAN	Whether the row represents the most recent or active version of account record (e.g., Y, N)
table_dim	table_dim_key (PK)	INTEGER	Unique surrogate key for the table in the dimension
table_dim	table_owner (NK)	STRING	The user who is responsible for the table
table_dim	table_name (NK)	STRING	The logical name of the data table
datamart_dim	datamart_dim_key (PK)	INTEGER	Unique surrogate key for the data mart
datamart_dim	datamart_database (NK)	STRING	Name of the database
datamart_dim	datamart_schema (NK)	STRING	Owner of the database
datamart_dim	datamart_name	STRING	Abbreviation for the associated data mart
datamart_dim	datamart_desc	STRING	Full name of the data mart where the table resides
datamart_dim	datamart_subject	STRING	Subject area of that data mart (e.g., finance, time-tracking, and students)
datamart_dim	datamart_environment	STRING	The database environment type (e.g., DEV, PROD, and TEST)
role_dim	role_dim_key (PK)	INTEGER	Unique surrogate key for the role
role_dim	role_name (NK)	STRING	Name of the role (all unique roles that exist across datamart_roles &

			oracle_role_privs)
role_dim	role_description	STRING	Description of the role (left blank if role was only found in oracle_role_privs & doesn't have an associated description in datamart_roles)
last_query_date_dim	last_query_date_dim_key (PK)	INTEGER	Unique surrogate key for each date
last_query_date_dim	last_query_date (NK)	DATE	Full calendar date
last_query_date_dim	last_query_year	INTEGER	The year part of the full calendar date
last_query_date_dim	last_query_month	INTEGER	The month part of the full calendar date
last_query_date_dim	last_query_day_no	INTEGER	The day part of full calendar date
last_query_date_dim	last_query_day_of_week	VARCHAR	The day in the week
access_usage_fact	account_dim_key (FK)	INTEGER	Unique surrogate key for the account in the dimension
access_usage_fact	datamart_dim_key (FK)	INTEGER	Unique surrogate key for the data mart
access_usage_fact	role_dim_key (FK)	INTEGER	Unique surrogate key for the role
access_usage_fact	table_dim_key (FK)	INTEGER	Unique surrogate key for the table in the dimension
access_usage_fact	last_query_date_dim_key (FK)	INTEGER	Unique surrogate key for each date
access_usage_fact	acct_name (NK)	STRING	Account name that can be either netid or application username?
access_usage_fact	database_name (NK)	STRING	Name of the database
access_usage_fact	database_role (NK)	STRING	Role the owner of the database
access_usage_fact	table_owner (NK)	STRING	The user who is responsible for the table
access_usage_fact	table_name (NK)	STRING	The logical name of the data table

access_usage_fact	table_privilege (DD)	BOOLEAN	Whether the privilege can be passed to others
access_usage_fact	total_query_count	INTEGER	Number of accesses/queries performed for the given account

ETL Source-to-Target Mapping

Source Table	Source Column	Target Table	Target Column	Transformation Logic	Notes
ORACLE _USERS	USERNAME	ACCOUNT _DIM	ACCT_NAME	Group by username	Natural key
ORACLE _USERS	USERNAME	ACCOUNT _DIM	ACCT _TYPE	Personal if NETID, else Application	Derived field
CU_PERSON	FIRST_NAME	ACCOUNT _DIM	ACCT_FNAME		
CU_PERSON	LAST_NAME	ACCOUNT _DIM	ACCT_LNAM E		
CU_PERSON	ACTIVE	ACCOUNT _DIM	ACCT_OWNE R_EMP _STATUS	Map A/I/R/N to Active status	
CU_PERSON	UNIT_NAME	ACCOUNT _DIM	ACCT__OWNE R_UNIT		
CU_PERSON	DEPARTMENT_ NAME	ACCOUNT _DIM	ACCT_OWNE R_DEPARTME NT		
APPID _OWNERS	ACCOUNT_NA ME	ACCOUNT _DIM	ACCT_USERN AME	Join for application accounts	
APPID _OWNERS	OWNER_NETID	ACCOUNT _DIM	DSS_START_D ATE	From SCD2 effective date	Derived field
-	-	ACCOUNT _DIM	DSS_END_DA TE	Default 9999-12-31 for current record	Derived field

-	-	ACCOUNT_DIM	DSS_CURRENT_FLAG	'Y' for current row	Derived field
ORACLE_ROLES_TO_TABLES	OWNER	TABLE_DIM	TABLE_OWNER		Natural key
ORACLE_ROLES_TO_TABLES	TABLE_NAME	TABLE_DIM	TABLE_NAME		Natural key
DATAMART_MAP	DATABASE_NAME	DATAMART_DIM	DATAMART_DATABASE		Natural key
DATAMART_MAP	OWNER	DATAMART_DIM	DATAMART_SCHEMA		Natural key
DATAMART_MAP	DATAMART_ABREV	DATAMART_DIM	DATAMART_NAME	Join on OWNER and DATABASE_NAME	
DATAMART_MAP	DATAMART_NAME	DATAMART_DIM	DATAMART_DESC		
DATAMART_MAP	DATAMART_SUBJECT	DATAMART_DIM	DATAMART_SUBJECT		
DATAMART_MAP	ENVIRONMENT	DATAMART_DIM	DATAMART_ENVIRONMENT	DEV/TEST/PROD	
ORACLE_ROLE_PRIVS	ROLE_NAME	ROLE_DIM	ROLE_NAME	Outer join with DATAMART_ROLES to include missing roles	Natural key
DATAMART_ROLES	ROLE_DESCRIPTION	ROLE_DIM	ROLE_DESC		
ORACLE_ROLES_TO_TABLES	PRIVILEGE	PRIVILEGE_DIM	PRIVILEGE_TYPE		
ORACLE_TABLE_AUDIT	ACTION_NAME	ACTION_DIM	ACTION_NAME	Uppercase	Natural key

ORACLE_TABLE_AUDIT	QUERY_DATE	DATE_DIM	YEAR / MONTH / DAY	Extract from QUERY_DATE	Derived field
ORACLE_TABLE_AUDIT	QUERY_DATE	ACCESS_USAGE_FACT	LAST_QUERY_DATE_DIM_KEY	Lookup from LAST_QUERY_DATE_DIM	Foreign key
ORACLE_ROLES_TO_TABLES	OWNER, TABLE_NAME	ACCESS_USAGE_FACT	TABLE_DIM_KEY	Lookup from TABLE_DIM	Foreign key
ORACLE_ROLE_PRIVS	ROLE_NAME	ACCESS_USAGE_FACT	ROLE_DIM_KEY	Lookup from ROLE_DIM	Foreign key
ORACLE_ROLES_TO_TABLES	DATABASE_NAME	ACTIVITY_FACT	DATAMART_DIM_KEY	Lookup from DATAMART_DIM	Foreign key
ORACLE_TABLE_AUDIT	USERNAME	ACTIVITY_FACT	ACCT_DIM_KEY	Lookup from ACCOUNT_DIM	Foreign key
ORACLE_TABLE_AUDIT	OBJ_NAME	ACTIVITY_FACT	TABLE_DIM_KEY	Lookup from TABLE_DIM	Foreign key
ORACLE_TABLE_AUDIT	PRIVILEGE, ACTION_NAME	ACCESS_USAGE_FACT	TABLE_PRIVILEGE	Lookup from TABLE_DIM	Natural key
ORACLE_TABLE_AUDIT	QUERY_DATE	ACCESS_USAGE_FACT	LAST_QUERY_DATE	MAX(query_date)	Derived field
ORACLE_TABLE_AUDIT	QUERY_COUNT	ACCESS_USAGE_FACT	TOTAL_QUERY_COUNT	Sum by account/date/table/action	Derived field

Assumptions and Questions

Requirements Clarification

Assumptions

1. The populated dimensional model in the team's database accurately reflects the requirements defined in Milestone 1, with ACCESS_USAGE_FACT continuing to represent both access grants and usage events at the same grain.

2. The unified fact table design supports simplified joins and consolidated reporting for IT Security, University Audit, and Data Stewards, aligning with the proof-of-concept BI dashboard requirements.

Questions

1. Should the BI dashboards differentiate *Access* and *Usage* metrics into two separate visualizations, or continue combining them in a single unified view for sponsors?

2. Are additional performance or compliance metrics (e.g., number of inactive accounts over time, trend of unused roles) expected for Milestone 3 deliverables?

Source Data Clarification

Assumptions

1. The source systems (Finance, Contributor Relations, and Student) expose reliable metadata on privileges, roles, and tables in a format compatible with the ETL mappings implemented in WhereScape.

Questions

1. How will discrepancies between ORACLE_USERS, ROLE_PRIVS, and TABLE_AUDIT sources be resolved when the same account has mismatched role or privilege information?

2. Are there expected latency or synchronization differences among these Oracle sources that could affect ETL load accuracy or dashboard timeliness?

Business Process Clarification

Assumptions

1. ETL jobs are configured to run on a scheduled basis (e.g., nightly or weekly) to maintain up-to-date dashboards and reflect current access privileges.

2. Changes to account employment status, department, and role assignments are propagated through nightly ETL refreshes using SCD Type 2 logic in WhereScape to maintain historical accuracy.

Questions

1. What is the expected turnaround time for reflecting changes (e.g., new hires, retired employees) in the BI system after updates occur in source systems?